

SmartBranch 360

Build a secure branch office network and prove it works

In one sentence: Design a small office network in Packet Tracer, make employee, guest, and server traffic work correctly, then write a Python tool that checks configs and helps you find mistakes.

Problem Statement

A company is opening a new branch office. The branch needs wired employees, guest Wi-Fi, an internal server, internet access, and secure management of network devices. Right now, there is no working design and no easy way to check whether the network is configured correctly.

Your team must design and build this branch network in Cisco Packet Tracer, test that traffic flows correctly, intentionally break the network to practice troubleshooting, and create a small Python assurance tool that validates the design.

MAIN COURSE Networking Essentials	TOOL Packet Tracer	PROGRAMMING Python	TEAM SIZE 2-3 students
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What You Must Build

Component	Requirement
Network topology	1 router, 2 switches, 1 wireless AP, 1 server, internet/cloud, 8 or more endpoints
VLANs	Employee, Guest, Server, Management (example: VLANs 10, 20, 30, 99)
IP plan	Separate subnet per VLAN (example: 10.10.10.0/24, 10.10.20.0/24, etc.)
Services	DHCP per VLAN, DNS reachability, NAT for internet, routing between VLANs
Security	Guest blocked from server VLAN; SSH allowed only from management VLAN
Python tool	Reads a plan or show outputs and reports likely configuration problems

Step-by-Step Workflow

- Plan the network:** Draw a topology diagram. Create a VLAN and IP address table. Decide which devices belong to each VLAN.
- Build in Packet Tracer:** Add and connect devices. Configure VLANs, trunks, router sub-interfaces, DHCP, NAT, and ACLs.
- Test normal operation:** Verify employee PCs reach the server and internet. Verify guest PCs reach internet only. Verify only the management PC can SSH to devices.
- Create fault scenarios:** Break the network on purpose at least 5 times (wrong gateway, missing VLAN on trunk, bad DHCP, ACL blocking DNS, NAT issue, etc.).
- Build the Python checker:** Write a script that validates IP/VLAN plans and/or show-command output. It should print clear findings such as missing VLAN, wrong gateway, or ACL blocking traffic.
- Record the demo:** Show the working network, inject one fault, run your Python tool, fix the issue in Packet Tracer, and verify with ping or traceroute.

Deliverables

Item	What to submit
Packet Tracer file	SmartBranch360.pkt with labeled VLANs and saved device configs
Design document	Topology diagram, VLAN/IP table, security rules, troubleshooting notes
Requirement file	Simple YAML or spreadsheet with site name, VLANs, and subnets
Python tool	Script plus sample validation report output
Fault cards	At least 5 written fault scenarios with symptom, root cause, and fix
Demo video	5 to 10 minutes showing build, fault, diagnosis, fix, and verification

How Your Work Will Be Checked

Check	Pass condition
Employee traffic	Employee PC gets DHCP, reaches server, and reaches internet
Guest isolation	Guest PC reaches internet but not server or management network
Secure management	Only management host can SSH to network devices
Troubleshooting	Each injected fault is diagnosed with evidence and fixed successfully
Automation	Python tool produces useful, readable validation output

Example Python finding

VLAN 20 missing on trunk SW1-F0/1 | Symptom: Guest PC has no connectivity | Suggested fix: Add VLAN 20 to allowed trunk list